

IMGD 4099 VFX Portfolio Debrief

EmberGen, LiquiGen, & GeoGen

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Overview

I looked into three beginner level VFX workflows based on EmberGen, LiquiGen, and GeoGen. To begin with, at the start of the term, I explored the basics of Unreal Engine tutorial videos. However, after learning about the capabilities of EmberGen, I became interested in creating VFX independently, using separate applications for simulations. After discovering EmberGen and its alternatives, I chose to experiment with visual effects applications and create three effects I deemed visually interesting: the nuclear explosion with a mushroom cloud created in EmberGen, a wave crashing down in LiquiGen, and a mountain crevasse terrain in GeoGen. The ultimate goal was to see how different effects were designed and to learn the basics of each one of these softwares.

EmberGen Nuke Explosion Effect

To start with, the first effect I created was the nuclear explosion effect. It was focused on generating an interesting mushroom cloud composed of various types of elements: fire, smoke, and ground particles to show a significant and impressive explosion. One of the main issues in creating this effect was adjusting the overall size and structure to make it look like an explosion rather than a fire. Moreover, another problem was making the ground particles more realistic. Another problem was the mushroom cloud itself, specifically the column part. In my version, the mushroom cloud “column” was fire dominant, and I spent many hours trying to figure out how to make that fire transition to smoke while maintaining the fire effect up at the top of the mushroom cloud at the same time. Unfortunately, I wasn’t able to solve this problem, but I learned a lot in trying to do so.

LiquiGen Crashing Wave Effect

The second effect I explored and created was a beach wave sequence. The goal here was to work on creating realistic water behavior and the way the liquid collides with itself. Unlike EmberGen, LiquiGen required me to concentrate more on flow, movement, and clarity of the fluid rather than smoke density or flame behaviors. Thus, I decided to try creating a simple crashing wave water effect that would involve water movements. This task required me to focus on timing and motion since even such an ordinary element as a wave should have a specific direction of movement and a regular rhythm to be realistic.

GeoGen Mountain Crevasse Effect

Another effect I produced used the terrain with fog added, specifically a mountain crevasse terrain. This effect differed from previous effects in a way that it was not based on simulation, rather, the focus was on creating the scenery itself. This effect was to be created with the goal to generate a mountain terrain with a large crevasse, and adding the fog in order to increase depth. One of the first problems I faced was in understanding how exactly the lighting worked and what parameters could be controlled in this regard. With the help from tutorials and by “experimenting” with parameters, I managed to obtain the desired effect. In addition, the problem included the generation of a suitable color palette for the scene as well. Again, with a combination of tutorials and trial-and-error process, and by understanding how exactly color affects the final result, the “problem” was successfully addressed.

Challenges and Problem Solving

Across all three projects, the most problematic part was working with the software and trying to achieve certain effects through it in a conscious manner. Since my experience in producing visual effects was limited, most of the time, I had to resort to YouTube tutorials to help me learn the softwares better.

Reflection / What I Learned

On the whole, this project helped me understand a lot about the creation of visual effects compared to the beginning of the course. One of the surprises I got was how these simulation software tools I worked with are based on various physical parameters, such as fuel, temperature, oxygen, density, motion, flow. Before working with these three programs (EmberGen, LiquiGen, and GeoGen), I thought that softwares such as these would have much more parameters that were more “disconnected” from the simulation, in the way that I would have to basically edit and guide my simulation frame by frame. Instead, when I first used EmberGen, for example, it felt like I was actually creating a real explosion by choosing how much “fuel”, and “oxygen” I wanted it to have. As a beginner, at first I did feel very overwhelmed with these softwares. However, overall, creating these VFX was very rewarding and I learned so much from doing so, and I am interested in continuing to learn how to use JangaFX in my spare time to see what else I can simulate.